

1 version 1

$$\begin{aligned}
\min E(Q, T) &= \int_{t_1}^{t_F} \int_{\Omega} p(t) Q(x, t) \, dx dt + \frac{\gamma}{2} \int_{\Omega} |T(x, t_F) - T_{\text{fin}}(x)|^2 \, dx \\
\begin{cases} \rho c \dot{T}(x, t) = \nabla \cdot (k(x) \nabla T(x, t)) + Q(x, t) & \text{in } \Omega \times (0, t_F) \\ -k(x) \partial_{\nu} T(x, t) = h(x) (T(x, t) - T_{\text{out}}(x, t)) & \text{on } \partial\Omega \times [0, t_F] \\ T(x, 0) = T_{\text{ini}}(x) \end{cases} & \quad (\text{SE}) \\
Q \in U_{\text{ad}} = \{Q \in L^2(\Omega \times (0, t_F)) \mid 0 \leq Q(x, t) \leq Q_{\text{max}}(x) \text{ a.e.}\} & \\
\text{optional : } T(x, t_F) \geq T_{\text{fin}}(x) & \text{(optional if second part of E should be strict)}
\end{aligned}$$

where $T(x, t)$ temperature in K (Henning: I will use °C, but written K is better), $Q(x, t)$ heating output in $\frac{W}{m^3}$, $p(t)$ price, ρ density in $\frac{kg}{m^3}$, c heat capacity in $\frac{J}{kg \cdot K}$, $k(x)$ heat transfer coefficient in $\frac{W}{m \cdot K}$, $h(x)$ convective heat transfer coefficient in $\frac{W}{m^2 K}$, t_1 morning start time

Remember that $\partial_{\nu} T$ is the outer normal derivative with respect to x and any initial data (or end data) would have to satisfy the PDE.

In order to solve the (SE) we will use a more basic method, in which we separate between the spacial and temporal derivatives and first solve the PDE in space using approximations of the time derivatives and then the ODE in time or we try to decouple both variables as much as possible to get a linear system in block structure. This is done by Gridap automatically. Thus, we need the weak formulation only in space (so a semi-weak formulation of the whole system). Here we will give it in its weak form remarking that every time integral of the (WF) for the (SWF) would be replaced by "for all $t \in (0, t_F)$ ", because computing the derivative of the energy functional is easier this way without losing track of the t , and get with the test space $H_{\Gamma}^1(\Omega \times (0, t_F))$ where $\Gamma = \Omega \times \{0\}$

$$\begin{aligned}
\int_0^{t_F} \int_{\Omega} \rho c \dot{T}(x, t) \phi(x, t) \, dx dt &= \int_0^{t_F} \int_{\Omega} \nabla \cdot (k(x) \nabla T(x, t)) \phi(x, t) + Q(x, t) \phi(x, t) \, dx dt \\
&= \int_0^{t_F} \int_{\Omega} -k(x) \nabla T(x, t) \cdot \nabla \phi(x, t) + Q(x, t) \phi(x, t) \, dx dt + \int_0^{t_F} \int_{\partial\Omega} k(x) \partial_{\nu} T(y, t) \phi(x, t) \, dy dt \\
&= \int_0^{t_F} \int_{\Omega} -k(x) \nabla T(x, t) \cdot \nabla \phi(x, t) + Q(x, t) \phi(x, t) \, dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) (T(y, t) - T_{\text{out}}(y, t)) \phi(x, t) \, dy dt
\end{aligned}$$

giving us the weak formulation

$$\int_0^{t_F} \int_{\Omega} \rho c \dot{T}(x, t) \phi(x, t) \, dx dt = \int_0^{t_F} \int_{\Omega} -k(x) \nabla T(x, t) \cdot \nabla \phi(x, t) + Q(x, t) \phi(x, t) \, dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) (T(y, t) - T_{\text{out}}(y)) \phi(y, t) \, dy dt \quad (\text{WF})$$

and thus

$$\begin{aligned}
a_{\text{SE}}(T, \phi) &= \int_0^{t_F} \int_{\Omega} \rho c \dot{T}(x, t) \phi(x, t) + k(x) \nabla T(x, t) \cdot \nabla \phi(x, t) \, dx dt + \int_0^{t_F} \int_{\partial\Omega} h(y) T(y, t) \phi(y, t) \, dy dt, \\
l_{\text{SE}}(\phi) &= \int_0^{t_F} \int_{\Omega} Q(x, t) \phi(x, t) \, dx dt + \int_0^{t_F} \int_{\partial\Omega} h(y) T_{\text{out}}(y) \phi(y, t) \, dy dt
\end{aligned}$$

Henning: remember that each boundary integral could be rewritten with the Trace operator, might be helpful for the next part. When writing the (SWF) the open interval for $t \in (0, t_F)$ is important.

From the optimisation functional and the (WF) we get the Lagrangian as

$$L(T, q, W) = \int_{t_1}^{t_F} \int_{\Omega} p(t) Q(x, t) \, dx dt + \frac{\gamma}{2} \int_{\Omega} |T(x, t_F) - T_{\text{fin}}(x)|^2 \, dx + \int_0^{t_F} \int_{\Omega} \rho c \dot{T}(x, t) W(x, t) + k(x) \nabla T(x, t) \cdot \nabla W(x, t) - Q(x, t) W(x, t) \, dx dt + \int_0^{t_F} \int_{\partial\Omega} h(y) (T(y, t) - T_{\text{out}}(y)) W(y, t) \, dy dt$$

which gives us the weak formulation of the adjoint (AE) as (notice that the second term contains $\dot{\psi}$ which in the (SWF) would be zero and thus needs to be solved differently: in that case we are actually only looking for the derivative of the x dependent T , so that the derivative $\partial_{T(x)}T(x, t)$ does not give 1 but a time dependent function)

$$L_T(T, Q, W)\psi = \gamma \int_{\Omega} (T(x, t_F) - T_{\text{fin}}(x)) \psi(x, t_F) dx + \int_0^{t_F} \int_{\Omega} \rho c \dot{\psi}(x, t) W(x, t) + k(x) \nabla \psi(x, t) \cdot \nabla W(x, t) dx dt + \int_0^{t_F} \int_{\partial\Omega} h(y) \psi(y, t) W(y, t) dy dt$$

By Green for any $t \in (0, t_F)$ it is

$$\int_{\Omega} k(x) \nabla \psi(x, t) \cdot \nabla W(x, t) dx = - \int_{\Omega} \nabla \cdot (k(x) \nabla W(x, t)) \psi(x, t) dx + \int_{\partial\Omega} k(y) \partial_{\nu} W(y, t) \psi(y, t) dy$$

and from Green for the t variable (integration by parts) we get

$$\int_0^{t_F} \int_{\Omega} \rho c \dot{\psi}(x, t) W(x, t) dx dt = - \int_0^{t_F} \int_{\Omega} \rho c \psi(x, t) \dot{W}(x, t) dx dt + \left[\int_{\Omega} \rho c \psi(x, t) W(x, t) dx \right]_0^{t_F} = - \int_0^{t_F} \int_{\Omega} \rho c \psi(x, t) \dot{W}(x, t) dx dt + \int_{\Omega} \rho c \psi(x, t_F) W(x, t_F) dx$$

when assuming $W(x, 0) = 0$ a.e. (because W has to be able to operate as test function for (SE)) and thus $\psi \in H^1_{\Gamma}(\Omega \times (0, t_F))$. With these it is

$$L_T(T, Q, W)\psi = \gamma \int_{\Omega} (T(x, t_F) - T_{\text{fin}}(x)) \psi(x, t_F) dx - \int_0^{t_F} \int_{\Omega} \rho c \psi(x, t) \dot{W}(x, t) + \nabla \cdot (k(x) \nabla W(x, t)) \psi(x, t) dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) \psi(y, t) W(y, t) - k(y) \partial_{\nu} W(y, t) \psi(y, t) dy dt + \int_{\Omega} \rho c \psi(x, t_F) W(x, t_F) dx$$

giving the strong version

$$\begin{cases} \rho c \dot{W}(x, t) = -\nabla \cdot (k(x) \nabla W(x, t)) & \text{in } \Omega \times (0, t_F) \\ k(y) \partial_{\nu} W(y, t) = h(y) W(y, t) & \text{on } \partial\Omega \times [0, t_F] \\ W(x, t_F) = -\frac{\gamma}{\rho c} (T(x, t_F) - T_{\text{fin}}(x)) & \text{in } \Omega \end{cases} \quad (\text{AE})$$

Thus, we can see that the (AE) has to be solved backwards in time. The weak version is

$$\int_0^{t_F} \int_{\Omega} \rho c \psi(x, t) \dot{W}(x, t) - k(x) \nabla \psi(x, t) \cdot \nabla W(x, t) dx dt = - \int_0^{t_F} \int_{\partial\Omega} h(y) \psi(y, t) W(y, t) dy dt \quad (\text{WAE})$$

For the (AE) we have

$$a_{\text{AE}}(W, \psi) = \int_0^{t_F} \int_{\Omega} \rho c \psi(x, t) \dot{W}(x, t) - k(x) \nabla \psi(x, t) \cdot \nabla W(x, t) dx dt + \int_0^{t_F} \int_{\partial\Omega} h(y) \psi(y, t) W(y, t) dy dt \quad l_{\text{AE}}(\psi) = 0$$

Henning: Actually, we get the (AE) by the variation principle and not by differentiation, which does not fully make sense here.

2 Variation

In order to find the adjoint, we introduce a small variation δT of T in the respective space and look at the difference this variation makes on L (this is like computing $L(T_1, Q, W) - L(T_2, Q, W)$, remember that the adjoint is a measure for the sensitivity of the functional with respect to T).

$$\delta_T L(T, Q, W) = \frac{\gamma}{2} \int_{\Omega} \delta_T |T(x, t_F) - T_{\text{fin}}(x)|^2 dx + \int_0^{t_F} \int_{\Omega} \rho c \delta T(x, t) W(x, t) - k(x) \nabla \delta T(x, t) \cdot \nabla W(x, t) dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) (\delta T(y, t) - T_{\text{out}}(y)) W(y, t) dy dt \quad (1)$$

$$= \gamma \int_{\Omega} (T(x, t_F) - T_{\text{fin}}(x)) \delta T(x, t_F) dx - \int_0^{t_F} \int_{\Omega} \rho c \delta T(x, t) \dot{W}(x, t) dx dt + \int_{\Omega} \rho c \delta T(x, t_F) W(x, t_F) dx - \int_0^{t_F} \int_{\Omega} k(x) \nabla \delta T(x, t) \cdot \nabla W(x, t) dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) (\delta T(y, t) - T_{\text{out}}(y)) W(y, t) dy dt \quad (2)$$

where we used the same as above and $\delta T(x, 0) = 0$ because of the initial value condition, which fixes the T s there and because the term is linear. Also for the first term we used Taylor giving us $x^2 = x_0^2 + 2x(x - x_0) + o((x - x_0)^2)$, giving us $\delta_x(x - a)^2 = (x - a)^2 - (x_0 - a)^2 = 2(x - a)(x - x_0) + o(x - x_0) = 2(x - a)\delta_x$.

Since the (AE) has to be solved backwards in time, which is not possible with the software we are using (although implementation would be just as is) we introduce $\widetilde{W}(x, t) = W(x, t_F - t)$. By the (AE) this function satisfies

$$\begin{cases} \rho c \dot{\widetilde{W}}(x, t) = \nabla \cdot (k(x) \nabla \widetilde{W}(x, t)) & \text{in } \Omega \times (0, t_F) \\ k(y) \partial_\nu \widetilde{W}(y, t) = h(y) \widetilde{W}(y, t) & \text{on } \partial\Omega \times [0, t_F] \\ \widetilde{W}(x, 0) = -\frac{\gamma}{\rho c} (T(x, t_F) - T_{\text{fin}}(x)) & \text{in } \Omega \end{cases} \quad (\text{AE}')$$

which can now be solved forward in time. The weak version is

$$\int_0^{t_F} \int_\Omega \rho c \psi(x, t) \dot{\widetilde{W}}(x, t) + k(x) \nabla \psi(x, t) \cdot \nabla \widetilde{W}(x, t) \, dx dt = \int_0^{t_F} \int_{\partial\Omega} h(y) \psi(y, t) \widetilde{W}(y, t) \, dy dt \quad (\text{WAE})$$

For the (AE') we have

$$a_{\text{AE}}(\widetilde{W}, \psi) = \int_0^{t_F} \int_\Omega \rho c \psi(x, t) \dot{\widetilde{W}}(x, t) + k(x) \nabla \psi(x, t) \cdot \nabla \widetilde{W}(x, t) \, dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) \psi(y, t) \widetilde{W}(y, t) \, dy dt \quad l_{\text{AE}}(\psi) = 0 \quad (3)$$

which then gives us the original adjoint as $W(x, t) = \widetilde{W}(x, t_F - t)$.

Now we introduce the solution operator S , which for a given Q gives the solution T of the (SE), so $T = S(Q)$. With this we can introduce the reduced energy functional

$$e(Q) = \int_{t_1}^{t_F} \int_\Omega p(t) Q(x, t) \, dx dt + \frac{\gamma}{2} \int_\Omega |S(Q)(x, t_F) - T_{\text{fin}}(x)|^2 dx \quad (4)$$

In order to find the gradient of e we differentiate it and find by noting that S is linear and therefore has derivative $S'(Q)h = Sh$

$$e'(Q)h = \int_{t_1}^{t_F} \int_\Omega p(t) h(x, t) \, dx dt + \gamma \int_\Omega (S(Q)(x, t_F) - T_{\text{fin}}(x)) Sh \, dx \quad (5)$$

$$= \langle p|h \rangle + \langle S(Q) - T_{\text{fin}} | Sh \rangle|_{t=t_F} = \langle p|h \rangle + \gamma \langle S^*(S(Q) - T_{\text{fin}}) | h \rangle|_{t=t_F} \equiv p - \rho c W \quad (6)$$

where S^* is the solution operator for (AE) and W is the solution

3 version 2

$$\min E(Q, T) = \int_{t_1}^{t_F} \int_\Omega (p(t) Q(x, t))^2 \, dx dt + \frac{\gamma}{2} \int_\Omega |T(x, t_F) - T_{\text{fin}}(x)|^2 dx \quad (7)$$

$$\begin{cases} \rho c \dot{T}(x, t) = \nabla \cdot (k(x) \nabla T(x, t)) + Q(x, t) & \text{in } \Omega \times (0, t_F) \\ -k(x) \partial_\nu T(x, t) = h(x) (T(x, t) - T_{\text{out}}(x, t)) & \text{on } \partial\Omega \times [0, t_F] \\ T(x, 0) = T_{\text{ini}}(x) \end{cases} \quad (\text{SE})$$

$$Q \in U_{\text{ad}} = \{Q \in L^2(\Omega \times (0, t_F)) \mid 0 \leq Q(x, t) \leq Q_{\text{max}}(x) \text{ a.e.}\} \quad (8)$$

$$\text{optional : } T(x, t_F) \geq T_{\text{fin}}(x) (\text{optional if second part of E should be strict}) \quad (9)$$

In this case everything stays the same but the Lagrangian is

$$L(T, q, W) = \int_{t_1}^{t_F} \int_\Omega (p(t) Q(x, t))^2 \, dx dt + \frac{\gamma}{2} \int_\Omega |T(x, t_F) - T_{\text{fin}}(x)|^2 dx + \int_0^{t_F} \int_\Omega \rho c \dot{T}(x, t) W(x, t) - k(x) \nabla T(x, t) \cdot \nabla W(x, t) - Q(x, t) W(x, t) \, dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) (T(y, t) - T_{\text{out}}(y)) W(y, t) \, dy dt \quad (10)$$

Now we introduce the solution operator S , which for a given Q gives the solution T of the (SE), so $T = S(Q)$. With this we can introduce the reduced energy functional

$$e(Q) = \int_{t_1}^{t_F} \int_{\Omega} (p(t)Q(x, t))^2 dx dt + \frac{\gamma}{2} \int_{\Omega} |S(Q)(x, t_F) - T_{\text{fin}}(x)|^2 dx \quad (11)$$

In order to find the gradient of e we differentiate it and find by noting that S is linear and therefore has derivative $S'(Q)h = Sh$

$$e'(Q)h = \int_{t_1}^{t_F} \int_{\Omega} 2p^2(t)Q(x, t)h(x, t) dx dt + \gamma \int_{\Omega} (S(Q)(x, t_F) - T_{\text{fin}}(x))Sh dx \quad (12)$$

$$= \langle 2p^2Q|h \rangle + \langle S(Q) - T_{\text{fin}}|Sh \rangle|_{t=t_F} = 2\langle p^2Q|h \rangle + \gamma \langle S^*(S(Q) - T_{\text{fin}})|h \rangle|_{t=t_F} \equiv 2p^2Q - \rho cW \quad (13)$$

Thus, without box constraints from the condition $e'(Q) = 0$ we would get $Q = -\frac{1}{2p^2}W$ and can use the projection of that onto the box for our case (see bang bang).

4 Optimal step size

We found out that the gradient of the reduced cost is $\nabla e(Q) = 2p^2Q - \rho cW$, so in the gradient descent algorithm we will go into the negative of that direction and we can actually compute the optimal step size. Consider a given iteration k and the computed control Q_k , state T_k and adjoint W_k . Also let $v_k = -\nabla e(Q) = -(2p^2Q - \rho cW)$, $I = [t_0, t_F]$ and define as the step-length-dependent function we want to minimise in the iteration

$$g(s) = e(Q_k + sv_k)$$

Then we can rewrite

$$g(s) = e(Q_k + sv_k) \quad (14)$$

$$= \|p(Q_k + sv_k)\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2} \|S(Q_k + sv_k) - T_{\text{textfin}}\|_{L^2(\cdot)}^2|_{t=t_F} \quad (15)$$

$$= \|pQ_k\|_{L^2(\Omega \times I)}^2 + 2sp^2Q_k, v_k|_{L^2(\Omega \times I)} + s^2\|pv_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|_{L^2(\cdot)}^2|_{t=t_F} + \gamma sT_k - T_{\text{fin}}, Sv_k|_{L^2(\cdot)}|_{t=t_F} + s^2\frac{\gamma}{2}\|Sv_k\|_{L^2(\cdot)}^2|_{t=t_F} \quad (16)$$

$$= s^2\left(\|pv_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|Sv_k\|_{L^2(\cdot)}^2|_{t=t_F}\right) + s\left(2p^2Q_k, v_k|_{L^2(\Omega \times I)} + \gamma T_k - T_{\text{fin}}, Sv_k|_{L^2(\cdot)}|_{t=t_F}\right) + 1\left(\|pQ_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|_{L^2(\cdot)}^2|_{t=t_F}\right) \quad (17)$$

$$= s^2\left(\|pv_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|Sv_k\|_{L^2(\cdot)}^2|_{t=t_F}\right) + s\left(2p^2Q_k, v_k|_{L^2(\Omega \times I)} + \gamma S^*(T_k - T_{\text{fin}}), v_k|_{L^2(\cdot)}|_{t=t_F}\right) + 1\left(\|pQ_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|_{L^2(\cdot)}^2|_{t=t_F}\right) \quad (18)$$

$$= s^2\left(\|pv_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|Sv_k\|_{L^2(\cdot)}^2|_{t=t_F}\right) + s\left(2p^2Q_k, v_k|_{L^2(\Omega \times I)} + \rho cW_k, v_k|_{L^2(\cdot)}|_{t=t_F}\right) + 1\left(\|pQ_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|_{L^2(\cdot)}^2|_{t=t_F}\right) \quad (19)$$

$$= s^2\left(\|pv_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|Sv_k\|_{L^2(\cdot)}^2|_{t=t_F}\right) - s\|v_k\|_{L^2(\Omega \times I)}^2 + 1\left(\|pQ_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|_{L^2(\cdot)}^2|_{t=t_F}\right) \quad (20)$$

This is a quadratic function in s and has the minimum in

$$s_{\min} = \frac{\|v_k\|_{L^2(\Omega \times I)}^2}{2\left(\|pv_k\|_{L^2(\Omega \times I)}^2 + \frac{\gamma}{2}\|Sv_k\|_{L^2(\cdot)}^2|_{t=t_F}\right)}$$

which requires to solve the (SE) again with input v_k instead of Q_k .

5 Projection Property

Theorem 1. Let V be a linear space (=vector space), $M \subseteq V$ a closed convex set, $f : V \rightarrow \mathbb{R}$ a strictly convex function and consider the unconstrained and constraint minimisation problems

$$\min_{x \in V} f(x) \quad \text{and} \quad \min_{x \in M} f(x).$$

Then each problem has at most one solution and if the unconstrained problem has a solution $x_V \in \arg\min_{x \in V} f(x)$ then the projection $\pi(x_V)$ is the solution to the constrained problem.

Proof. (i) Since M is arbitrary, proving that the constrained problem has at most one solution proves the claim also for the other case by setting $M = V$. Assuming there are two solution $x^1, x^2 \in M$, then $(x^1, x^2) \subseteq M$ by convexity of M and for every $x \in (x^1, x^2)$ by strict convexity of f it is $f(x) < \lambda f(x^1) + (1 - \lambda)f(x^2) = 1 \cdot \min_{y \in M} f(y)$, contradicting the minimality of x^1, x^2 .

(ii) Assume the minimum over M , denoted by x_M , would not be the projection. Then $f(x_V) < f(\pi(x_V))$, $f(x_V) < f(x_M)$ by (i) and $f(x_M) < f(\pi(x_V))$ and thus $f(x_V) < f(x_M) < f(\pi(x_V))$. x_M has to be in the boundary of M , otherwise we could as in (i) go a little bit into direction of x_V (better: the direction is $x_V - x_M$) and have smaller values. Connecting any two points on the triangle $T = \text{span}\{x_V, x_M, \pi(x_V)\}$ we thus get $f(x) < f(\pi(x_V))$ for every $x \in T \setminus \{\pi(x_V)\}$ $\square$

6 Remark: minimal time interval

If we consider the system without the boundary term, so without heat loss to the outside, then in order to be able to achieve an end temperature close to T_{fin} we need enough time, precisely

$$t = \frac{|\Omega|}{|Q_{\text{ad}}|} \frac{T_{\text{fin}} - T_{\text{ini}}}{\max Q}$$

much time, which corresponds to heating the room in average to the final temperature using maximum output all the time.

For small time intervalls we either get a bigger error in the corresponding term of the energy or need a big $\max Q$. Looking at the Theta-method with Y_n being any solution (of SE or AE) for only a given time t_n the next solution for t_{n+1} with the help of the right hand side extracted by looking at the time change of the system will be

$$Y_{n+1} = Y_n + \Delta t (\theta F(t_n, Y_n) + (1 - \theta) F(t_{n+1}, Y_{n+1}))$$

which gets solved by the Newton method iterating over k

$$Y_{n+1}^{k+1} = Y_{n+1}^k - \left(G'(Y_{n+1}^k) \right)^{-1} G(Y_{n+1}^k)$$

where

$$G(Y) = Y - \Delta t (\theta F(t_n, Y_n) + (1 - \theta) F(t_{n+1}, Y)) - Y_n, \quad G'(Y) = I - (1 - \theta) \Delta t \partial_Y F(t_{n+1}, Y)$$

7 Adjoint

Theorem 2. *The Adjoint solution operator satisfies*

$$\langle S^*(T - T_{\text{fin}}), h \rangle = \langle T - T_{\text{fin}}, S(h) \rangle$$

Proof. We remind about the weak forms (WSE) with test function W and (WAE) with test function T

$$\int_0^{t_F} \int_{\Omega} \rho c \dot{T}(x, t) W(x, t) \, dx dt = \int_0^{t_F} \int_{\Omega} -k(x) \nabla T(x, t) \cdot \nabla W(x, t) + Q(x, t) W(x, t) \, dx dt - \int_0^{t_F} \int_{\partial\Omega} h(y) (T(y, t) - T_{\text{out}}(y)) W(y, t) \, dy dt \quad (\text{WSE})$$

$$\int_0^{t_F} \int_{\Omega} \rho c \dot{T}(x, t) W(x, t) \, dx dt = - \int_0^{t_F} \int_{\Omega} k(x) \nabla T(x, t) \cdot \nabla W(x, t) \, dx dt - \gamma \int_{\Omega} (T(x, t_F) - T_{\text{fin}}(x)) T(x, t_F) \, dx - \int_0^{t_F} \int_{\partial\Omega} h(y) W(y, t) T(y, t) \, dy dt \quad (\text{WAE})$$

With these it is

$$\begin{aligned} \langle W, S(Q) \rangle &= \int_0^{t_F} \int_{\Omega} W(x, t) T(x, t) \, dx dt \\ &= \int_0^{t_F} \int_{\Omega} W(x, t) \left(T_{\text{ini}}(x) + \int_0^t \dot{T}(x, s) \, ds \right) \, dx dt \\ &= \int_0^{t_F} \int_{\Omega} W(x, t) T_{\text{ini}}(x) + \int_0^t \dot{T}(x, s) W(x, t) \, ds \, dx dt \\ &= \int_0^{t_F} \int_{\Omega} W(x, t) T_{\text{ini}}(x) + \int_0^t \dot{T}(x, s) W(x, t) \, ds \, dx dt \\ &= \end{aligned}$$

□

8 Other

- Remember the Courant-Friedrichs-Lewy (CFL) condition: If the numerical solution is screwed up it might be because space step length / time step length is too small. Honestly, I do not know the specific condition for that problem, it might even be reversed here, but this should be considered.
- Galerkin method or Ritz method - mention them and I think we use Galerkin
- Pontryagin's maximum principle??? Gives the adjoint of a system like ours, but it might be hard to transform